Introduction to Firebase

Database design

NIFR 2023

Topic: Firebase

Lectures overview

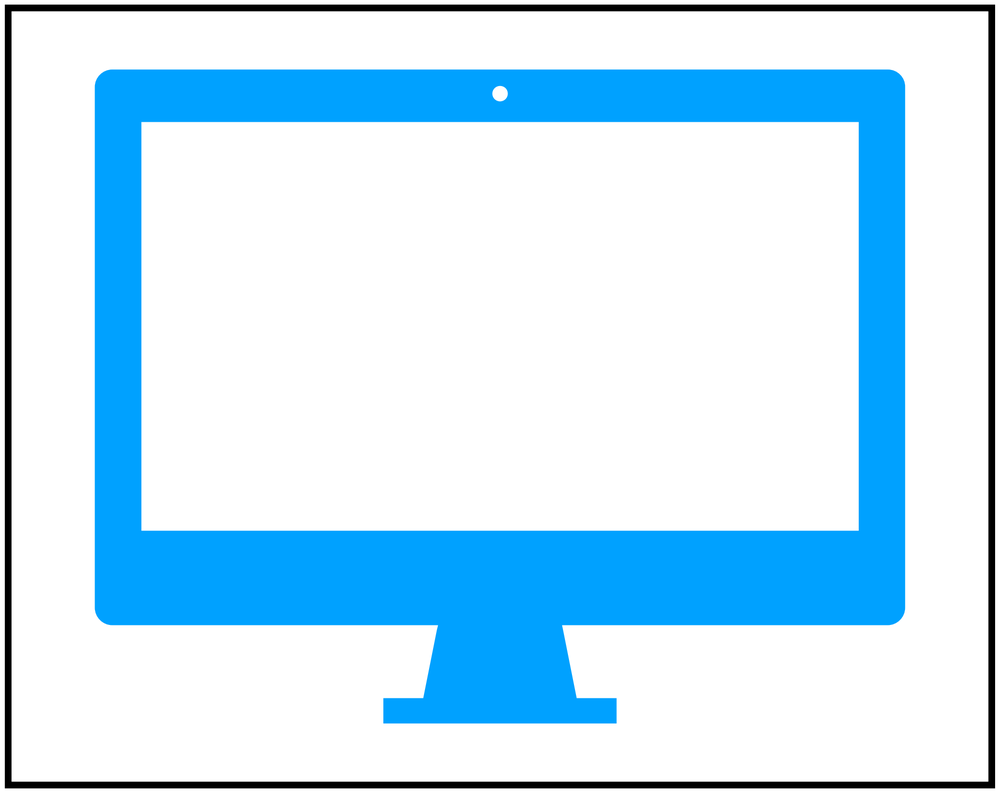
- 1. Firebase: Development setup
- 2. NoSQL & Cloud Firestore
- 3. (Online) Firebase Authentication
- 4. Firebase project: Javascript, HTML, css

Agenda

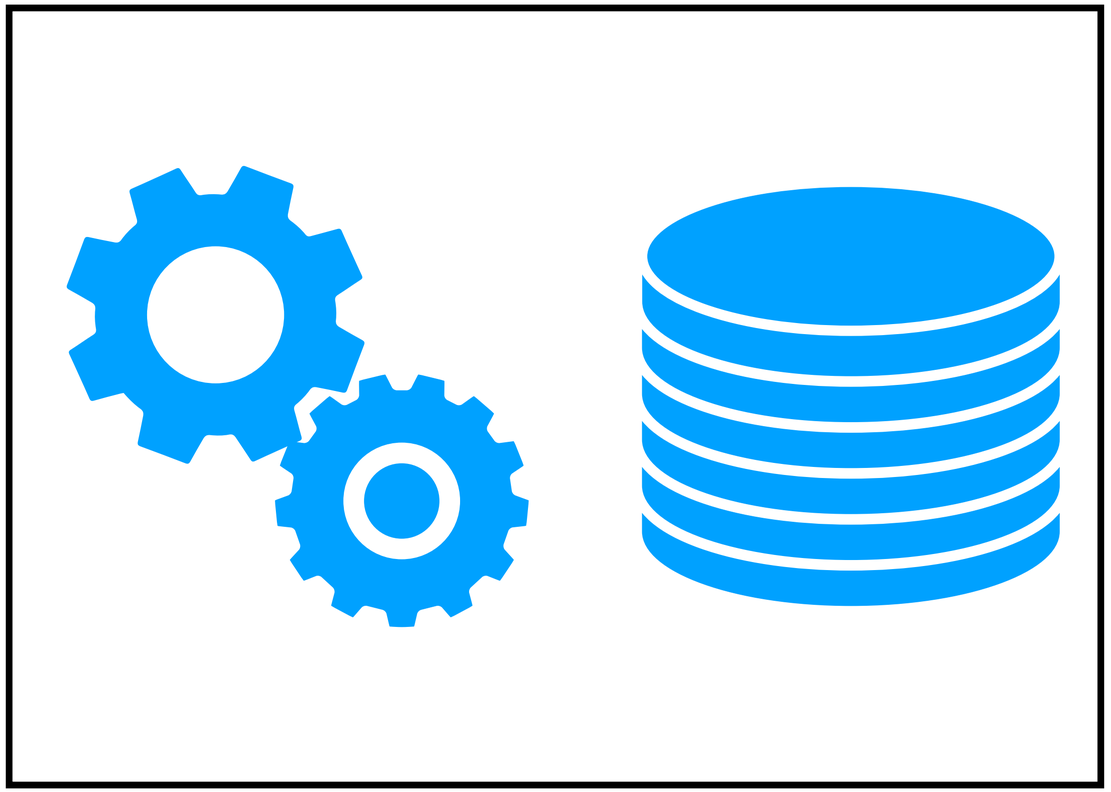
Database design: Firebase configuration

- Backend as a service (BAAS)
- Development environment
- Node Package Manager
- Webpack
- Firebase: Cloud Firestore

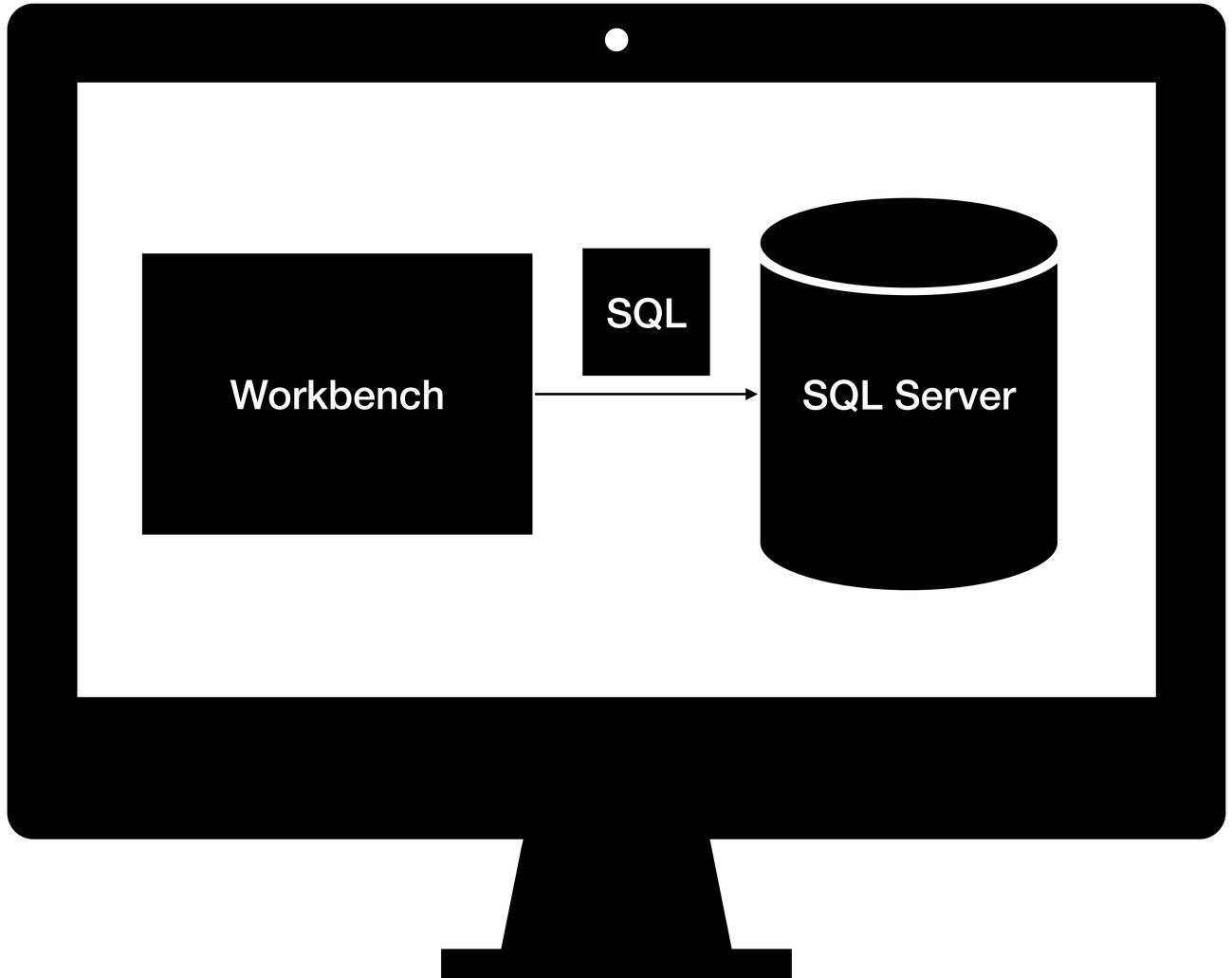
Front-end



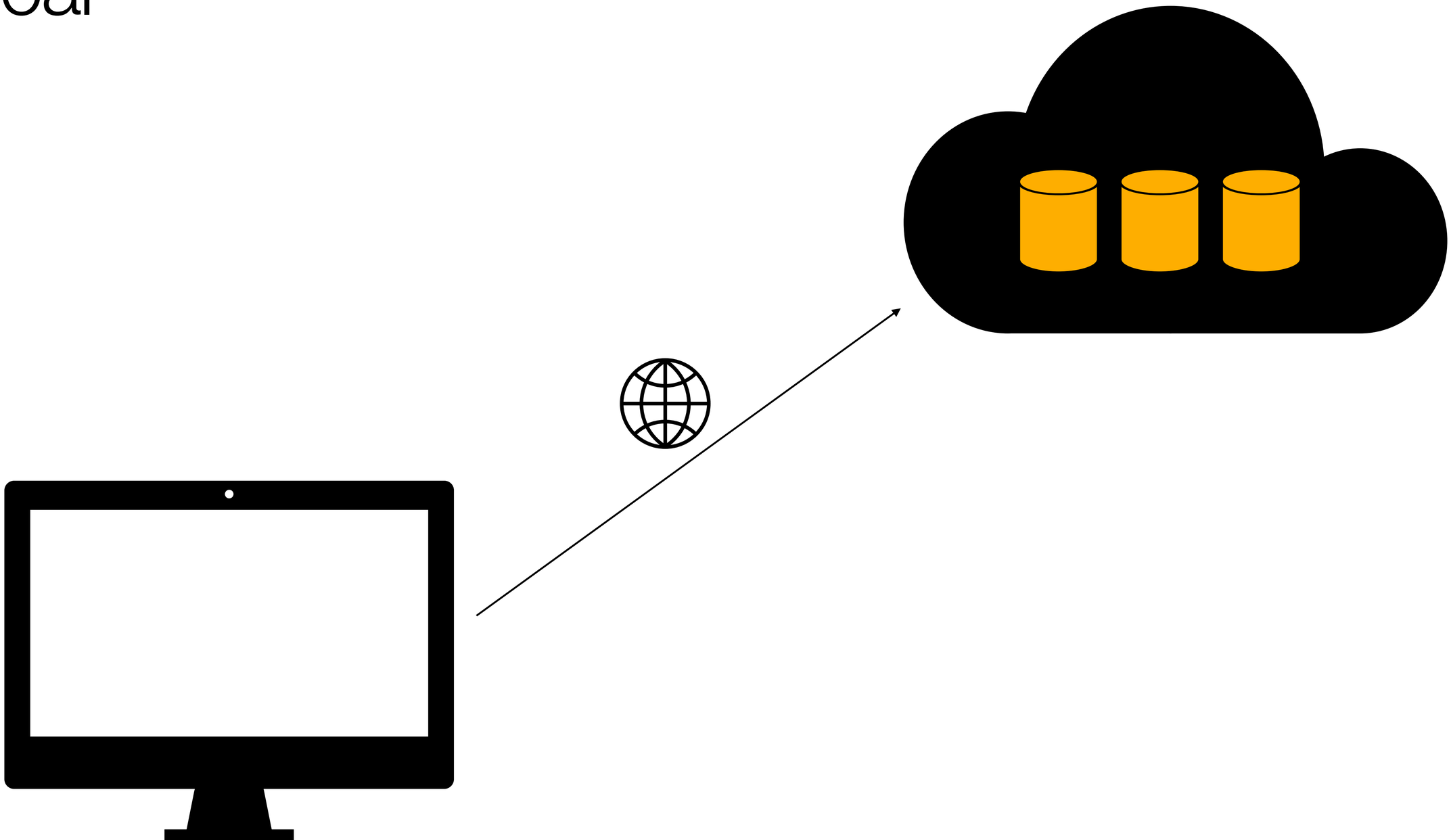
Backend



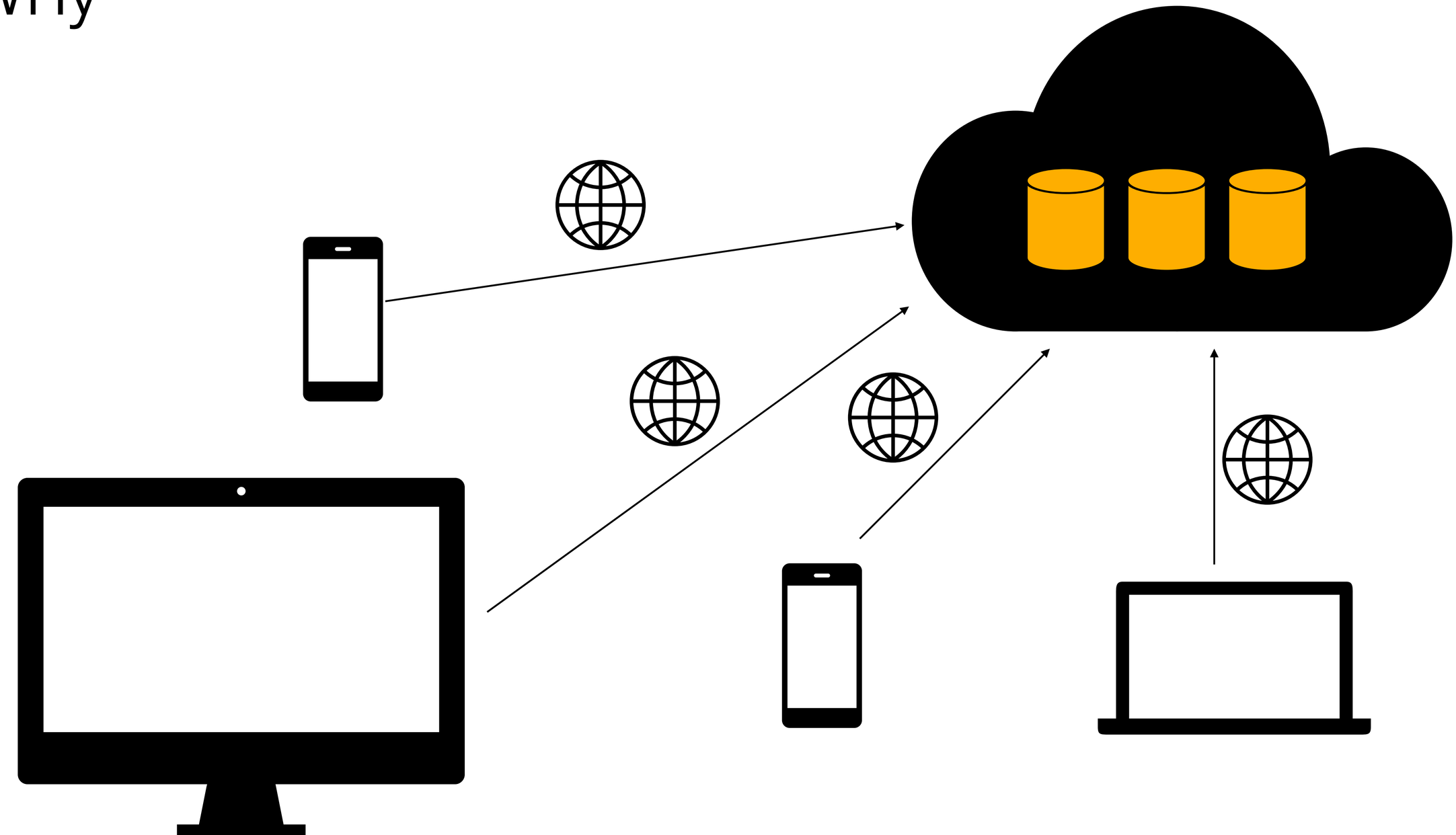
Local development



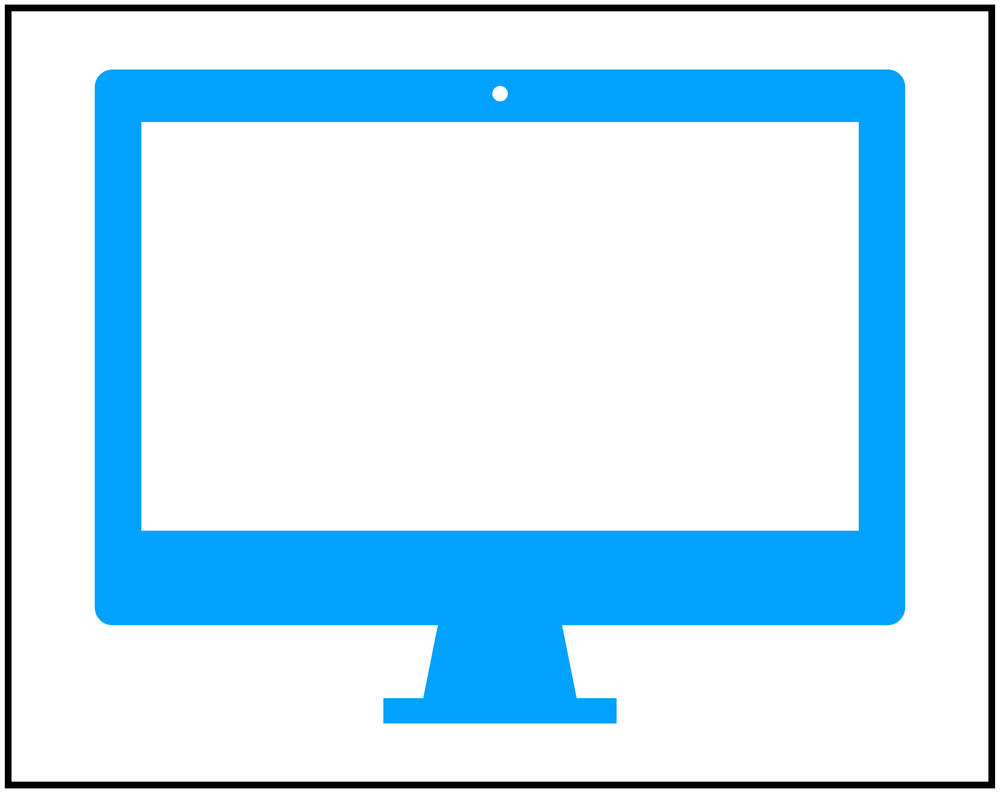
Goal



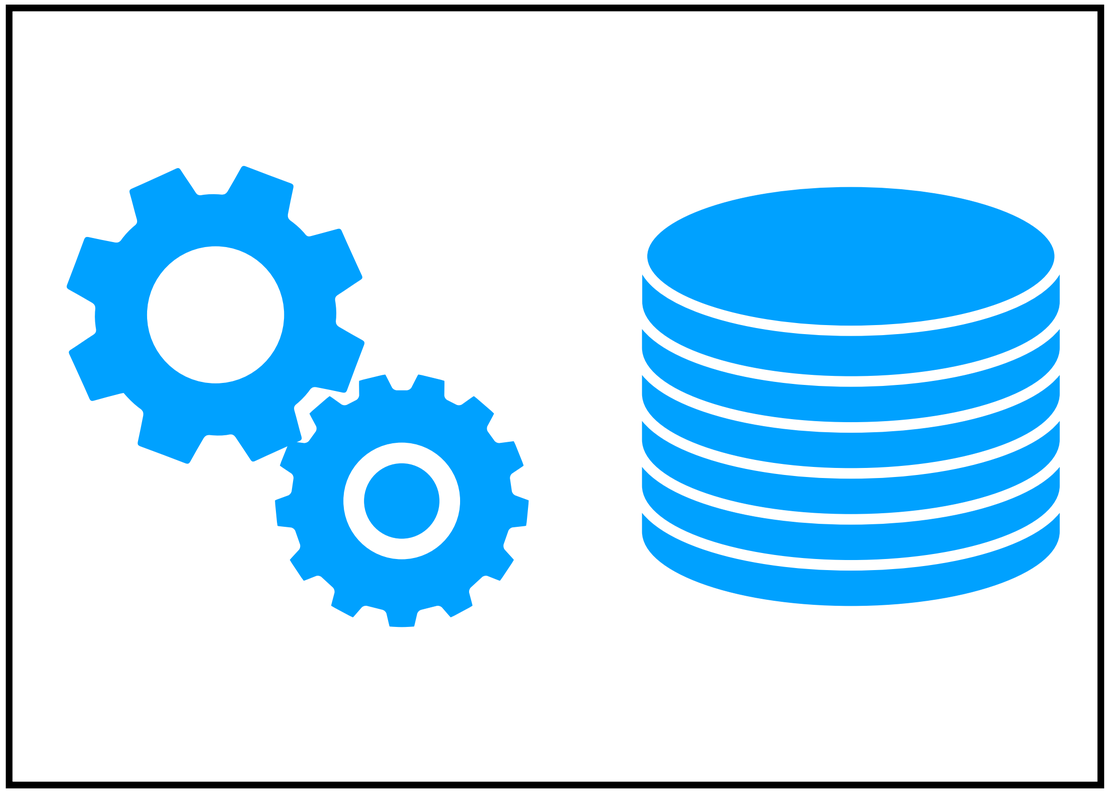
Why



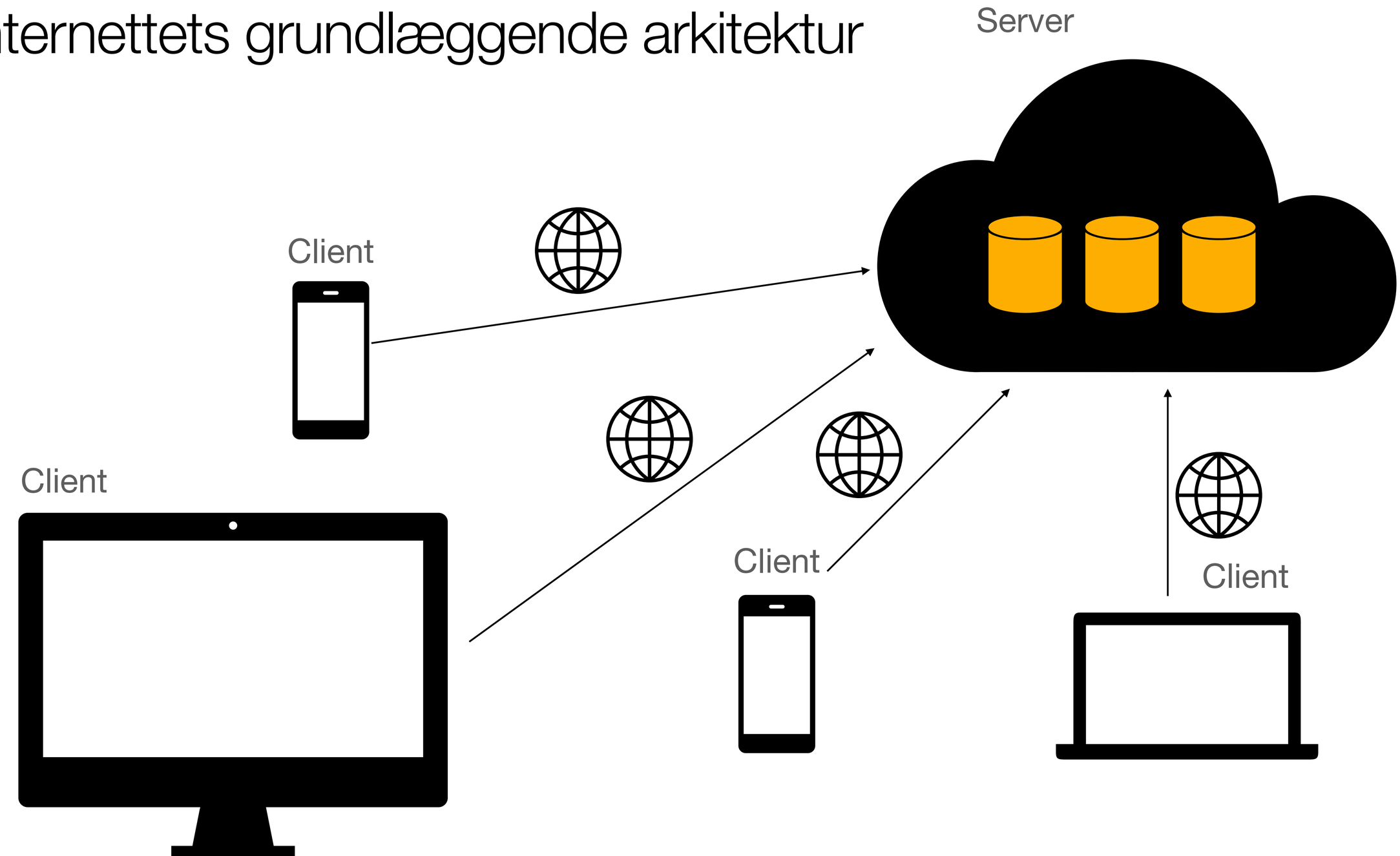
Client



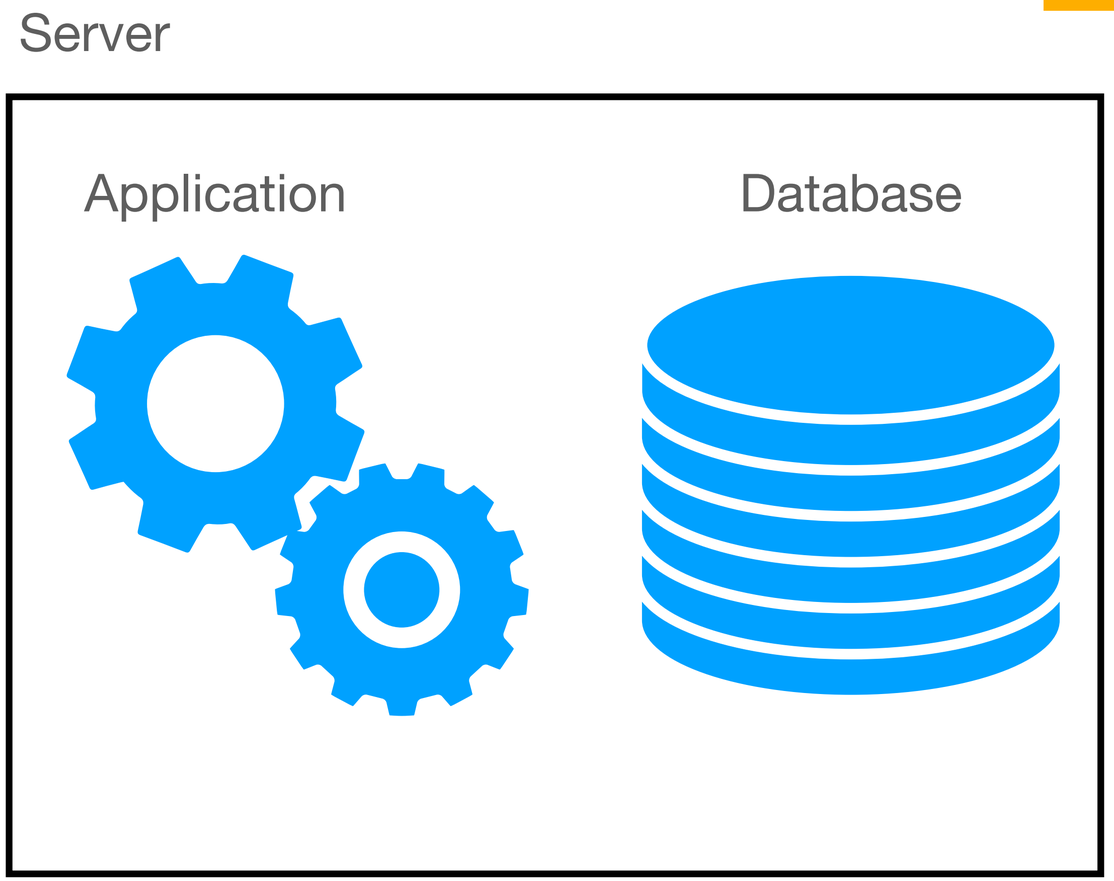
Server



Internettets grundlæggende arkitektur



MySQL Database server maintained by oneself



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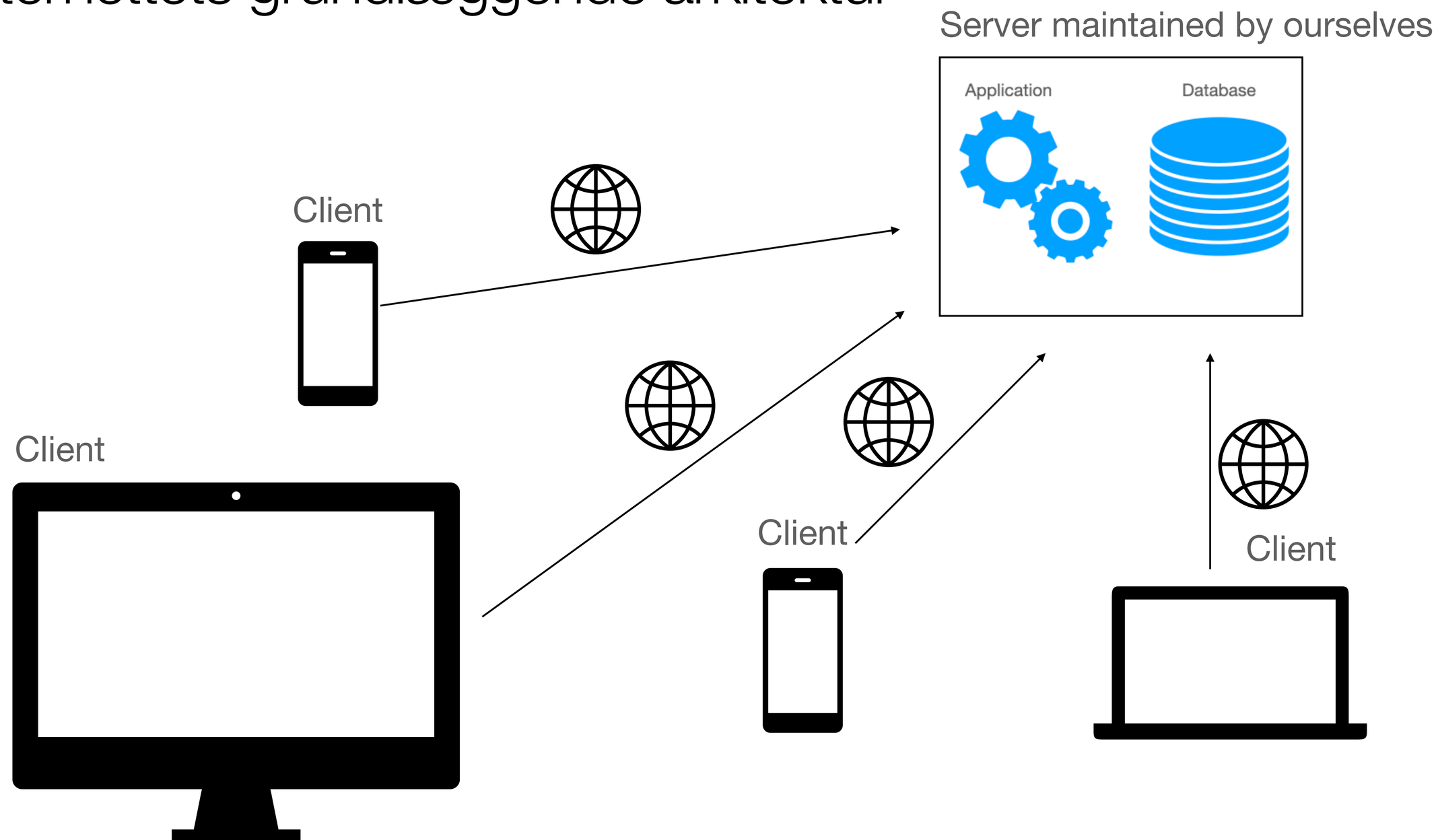
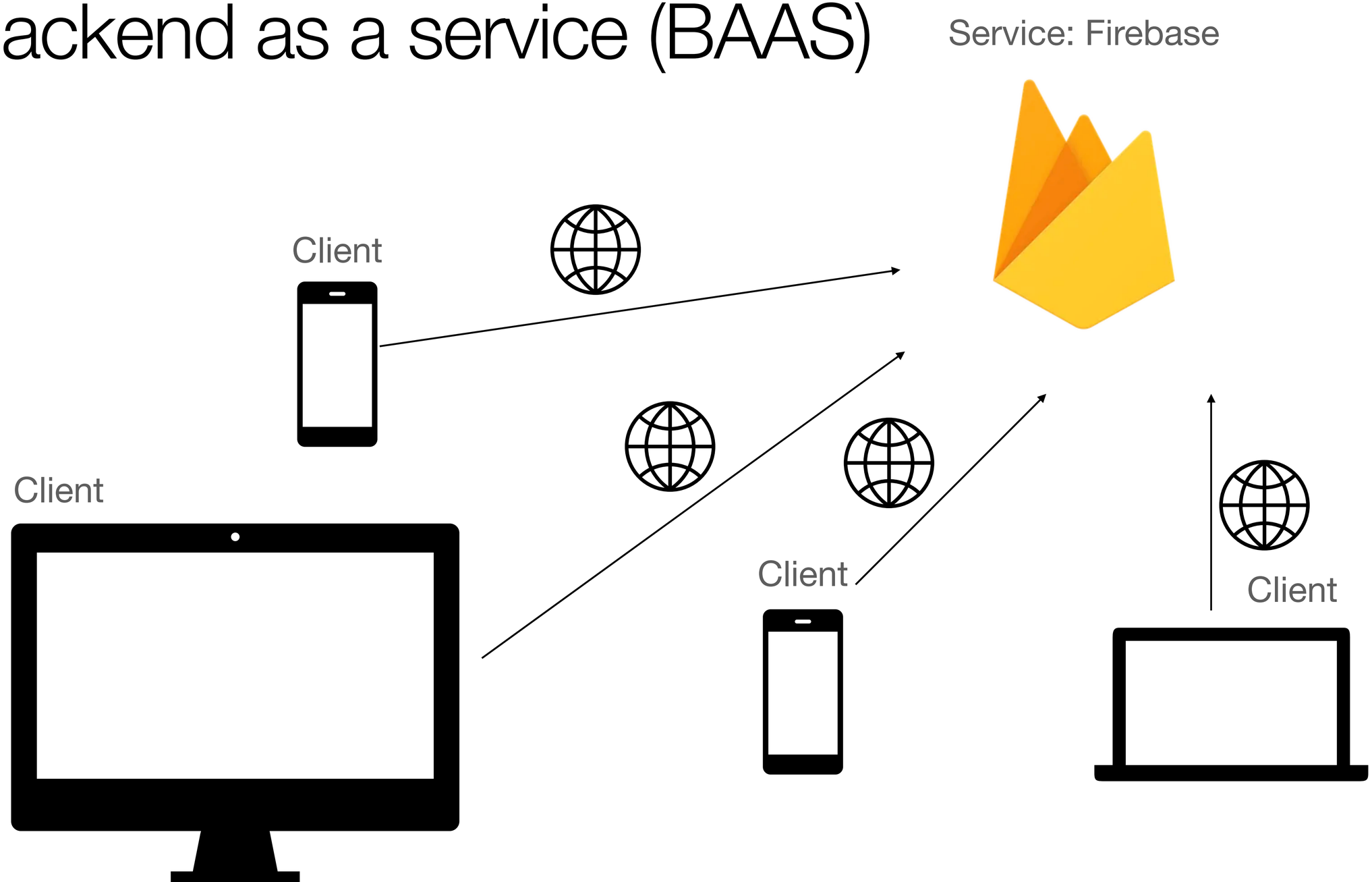


Backend as a service (BAAS)



Pricing by location

The following table lists pricing for reads, writes, deletes, and storage for each Firestore location:

Taiwan (asia-east1)

	Free quota per day	Price beyond the free quota (per unit)	Price unit
Document Reads	50,000	$0.0345	per 100,000 documents
Document Writes	20,000	$0.1042	per 100,000 documents
Document Deletes	20,000	$0.0115	per 100,000 documents
TTL Deletes	Not supported	$0.0115	per 100,000 documents
Stored Data	1 GiB storage	$0.1725	GiB/Month
PITR data	Not supported	$0.1725	GiB/Month
Backup data	Not supported	$0.035	GiB/Month
Restore operation	Not supported	$0.23	GiB

Backend as a service

Benefits

- Faster development - no/little backend development required
  - Less developers
  - Faster time to market
- Standardised backend development
  - Firebase/Azure as a “known” technology
- Serverless & scaleable
- Reduction of cost (in many cases)

Backend as a service

Why not?

- High volumes of data equals too high of a cost
- Predictable growth trajectories does not require sudden scaling
- Backend service providers are businesses and need to earn a margin

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October 19, 2022

Why we're leaving the cloud

The cloud excels at two ends of the spectrum, where only one end was ever relevant for us. The first end is when your application is so simple and low traffic that you really do save on complexity by starting with fully managed services. This is the shining path that Heroku forged, and the one that has since been paved by Render and others. It remains a fabulous way to get started when you have no customers, and it'll carry you quite far even once you start having some. (Then you'll later be faced with a Good Problem once the bills grow into the stratosphere as usage picks up, but that's a reasonable trade-off.)

The second is when your load is highly irregular. When you have wild swings or towering peaks in usage. When the baseline is a sliver of your largest needs. Or when you have no idea whether you need ten servers or a hundred. There's nothing like the cloud when that happens, like we learned when launching HEY, and suddenly 300,000 users signed up to try our service in three weeks instead of our forecast of 30,000 in six months.

https://world.hey.com/dhh/why-we-re-leaving-the-cloud-654b47e0

Setting up a development environment

Technologies

- Node package manager
- Webpack
- Firebase: Cloud Firestore

Node Package Manager (NPM)
Demo

Did I initiate a project? - Yes if you have a package.json file

```
{
  "name": "npm-demo",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "js-confetti": "^0.11.0"
  }
}
```

Development environment

NPM

- Initiate a new empty project
- Install NPM
- Find the terminal
- Write: npm init -y
- Verify if you have a package.json file

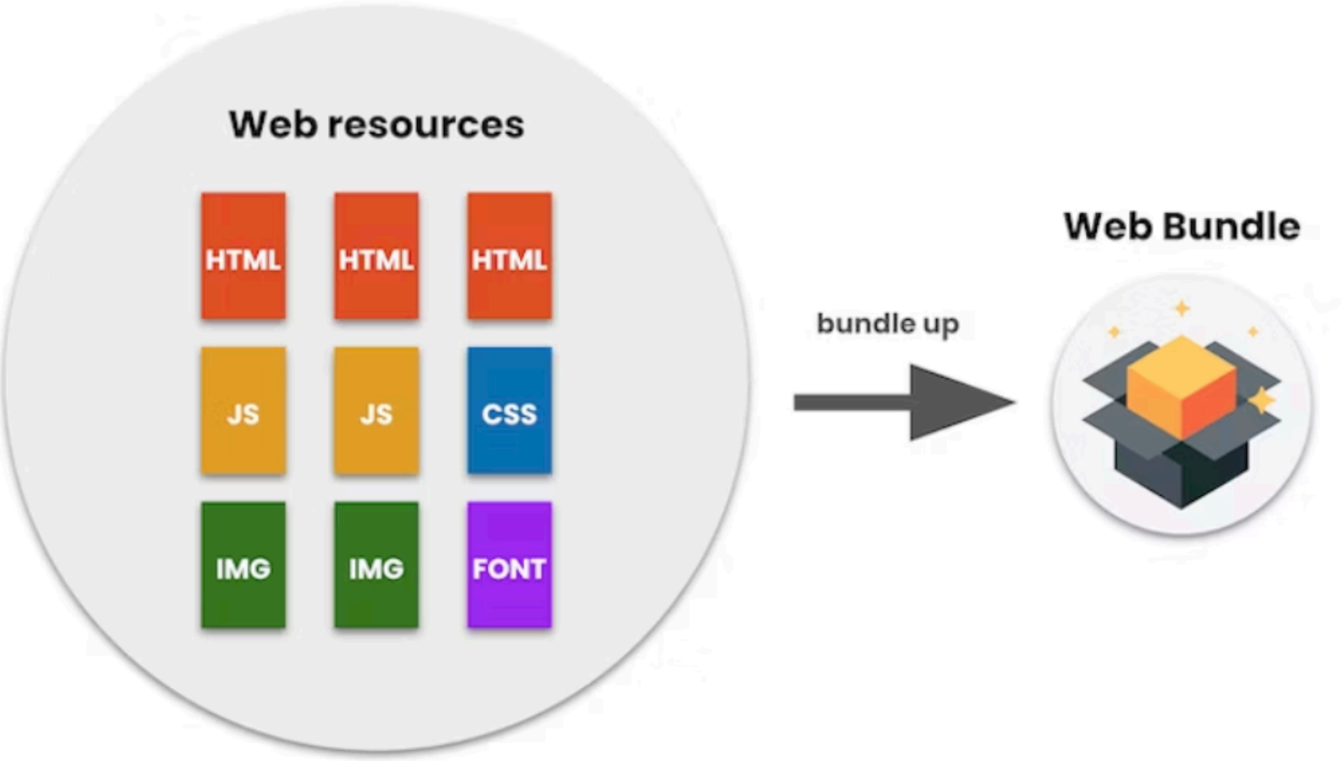
Javascript Module Bundler

Webpack

- Bundles application code with the packages from NPM
- Minifies/uglyfies code: .css, javascript, html
- Interprets between javascript supersets / frameworks: React, Typescript, Vue
- Reduces image size

- Makes source code ready to for the client

Web bundle



https://developer.chrome.com/docs/web-platform/web-bundles/

Minification

https://www.toptal.com/developers/cssminifier

```
body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
}

.container {
    width: 100%;
    max-width: 1200px;
    margin: 0 auto;
    padding: 20px;
}

.header {
    background-color: #333;
    color: #fff;
    padding: 10px;
}

.nav {
    list-style: none;
    display: flex;
}
```

Javascript Module Bundler

How to

- Install web pack
- Create new folder directory
- Insert content in webpack.config.js
- Change package.json
- Try your build: npm run build
- Gitbook:

Exercise 1

Firebase

Configuration

- Create a firebase project
- Register your application
- Install firebase with NPM

- Create a Cloud Firestore project
- Use test mode
- Select an EU location

- Locate your web config
- Create test-data in the Cloud Firestore console

Exercise 2 & 3